Name: _____

WORKSHEET

Set operations

1 Consider the set of natural numbers $\mathbb{N} = \{1,2,3,4,5\ldots\}$, the set of integers $\mathbb{Z} = \{\ldots -3,-2,-1,0,1,2,3,\ldots\}$, and the set of all real numbers $\mathbb{R} = (-\infty, \infty)$. Write the following:

a $\mathbb{N} \cup \mathbb{Z}$

b $\mathbb{N} \cap \mathbb{Z}$

c $\mathbb{Z} \cap \mathbb{R} \cap \mathbb{N}$

d $\mathbb{Z} \cup \mathbb{N} \cap \mathbb{R}$

2 Given $A = \{!,@,\#\}$, $B = \{@,\#,\$\}$, $C = \{\#,\$,\%\}$, find the following:

a $A \cup C$

b $B \cup C$

c $A \cap B$

d $A \cap C$

e $A \cup B \cup C$

f $A \cap B \cap C$

g $A \cap B \cup C$

h $C \cap A \cup B$

3 Given the universal set of elements $\{a,b,c,d,e,f,g,h,i,j\}$, and the sets $A = \{a,c,e,g,i\}$, $B = \{a,d,g,j\}$ and $C = \{c,d,f,h\}$, write the following:

a $A \cup B$

b $A \cap B$

c $\overline{C}$

d $\overline{B} \cup A$

e $A \cap \overline{C}$

f $\overline{B \cup C}$

g $\overline{A \cup B \cup C}$

h $\overline{B \cap C}$

i $\overline{A \cap B} \cup C$

j $\overline{A \cup \overline{B}}$

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4 True or False

a $\mathbb{Z} \subset \mathbb{Q}$

b $\mathbb{Q} \subset \mathbb{N}$

c $\mathbb{Z} \subset \mathbb{N}$

d $\mathbb{Q}$ is a proper subset of $\mathbb{R}$

e $\mathbb{Z}^+$ is a proper subset of $\mathbb{N}$

f $\mathbb{Z}^- \cap \mathbb{N} \quad \varnothing$

g $\sqrt{2} \subset \mathbb{Q}$

h $0.517 \subset \mathbb{Q}$

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Answers

1 **a** $\mathbb{Z}$
b $\mathbb{N}$
c $\mathbb{N}$
d $\mathbb{Z}$

2 **a** $\{!,@,\#,\$,\%\}$
b $\{@,\#,\$,\%\}$
c $\{@,\#\}$
d $\{\#\}$
e $\{!,@,\#,\$,\%\}$
f $\{\#\}$
g $\{@,\#,\$,\%\}$
h $\{@,\#,\$\}$

3 **a** $\{a,c,d,e,g,i,j\}$
b $\{a,g\}$
c $\{a,b,e,g,i,j\}$
d $\{a,b,c,e,f,g,h,i\}$
e $\{a,e,g,i\}$
f $\{b,e,i\}$
g $\{b\}$
h $\{a,b,c,e,f,g,h,i,j\}$
i $\{b,c,d,e,f,h,i,j\}$
j $\{d,j\}$

4 **a** T
b F
c F
d T
e F
f F
g F
h T

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